Chapter 1

Literature Review

1.1 Theoretical Motivations and Background

At the broadest level, this thesis seeks to contribute to an understanding of gender from a trans and queer critical perspective. Following the revolutionary work of Judith Butler (1990), gender scholarship has come to regard gender as performative. Most simply, this means that gender does not exist outside of culture and society, and is constantly constructed and reconstructed by social behavior. Linguistic behavior has always played a major role in performativity; Butler borrowed the term from linguist J. L Austin (1962), who noticed that by using language we do not simply make statements about the world, we affect and shape it. Butler adapted and revised the theory, redeveloping it on a sophisticated social-theoretical background Henry and Hall (2025, January 1). This renewed the interest of sociolinguists and linguistic anthropologists in performativity, especially those in the study of language and gender Bucholtz (2014) and Henry and Hall (2025, January 1). These scholars would come to connect performativity with the sophisticated theorizations of the social and cultural dynamics of language use they had already developed, primarily their models of semiotics (meaning-making) and indexicality Henry and Hall (2025, January 1). Indexicality is a particular dynamic within semiotics, so I will return to it after a discussion of the recent model of semiotics developed by Gal and Irvine (2019).

In Gal and Irvine's model, individuals make meaning of their world through conjectures. Drawing upon their knowledge and understanding of the world, they recognize a phenomenon and conjecture that it may be a sign; it may be meaningful. They conjecture that that sign signals (means or represents) a particular object, its meaning. Essential to note here is that these conjectures constitute the sign-object relationship; nothing is a sign in itself, independent of a perciever conjecturing that it is a sign and it signals an object (Gal and Irvine 2019, p. 89). Also important is that these conjectures are based on the perciever's knowledge and understanding of the world, and which are constructed and mediated by their cultures, communities, and individual social identities– their lived experience (Gal and Irvine 2019; see also Calder 2021). Part of a conjecture is its "grounds": the

reasoning the conjecture provides as to why its phenomenon is a sign and why it signals this object. One of these grounds, often centered in sociolinguistics, is termed 'indexical'. A conjecture with indexical grounds claims that a phenomenon is a sign of a particular object because the sign is understood to co-occur with, point to, or be caused by its object (Gal and Irvine 2019, p. 96).

Sociolinguistics often focuses on finding correlations between differences in language use (eg. pronunciation, grammar, word choice) and differences in social identity, persona, or context Eckert (2012). Particular linguistic forms are then said to index the social identities of those who use them, on the grounds of this relationship of co-occurence Eckert (2012). Termed by Silverstein (2023) the "social-indexical performativity of language-in-use" (p. 17), this sort of semiotic is a mechanism of gender performativity (Henry and Hall 2025, January 1): if a speaker uses a particular linguistic form, and the listener believes that form to be commonly used by those of a particular gender identity and thus conjectures that linguistic behavior as indexing that speaker's gender, this socioculturally-mediated perceiving act has performatively constructed their gender. That there are two actors in play here, a speaking subject and a listening subject, led Bucholtz and Hall (2004), Bucholtz and Hall (2005), and Bucholtz and Hall (2010, January 31) to understand to the performative construction of gender (and identity categories more broadly) as an intersubjective process.

Bucholtz and Hall characterize this intersubjective process as a negotiation (eg. Bucholtz and Hall 2005, p. 606) between the indexical goals of the speaker and the indexical goals of the listener. Although it is possible that a listener may construe an indexical meaning of a linguistic form used by a speaker who has no knowledge of, or intent to make use of, that form's indexicality, agentivity on the part of the speaker is common. Speakers have interactional goals, and hope that their behavior will be construed as indexical of those goals, whether it be related to identity, eg. presenting feminine, or other aspects such as persona and stance, eg. seeming like a 'diva' (see Podesva 2007). This agentic view arose principally in reaction to earlier models which stripped speakers of agency and individuality, instead understanding their linguistic behavior only to index their static, essentialized identities (Eckert 2012). In practice, this turn reframes "she speaks this way because she is a woman" as "she chooses this form of speech to construct a feminine identity." Acknowledging that individuals have choices about how to present themself and granting them authority over their self-identification is also a move towards alignment with queer and trans understandings of identity; trans people in the US often consider gender identity to be self-determined (see Zimman 2017), rather than a consequence of bodily characteristics. Despite the value of emphasizing the speaker's goals to legitimize their identities and agency, it is nonetheless the case that the listener is the other side of this negotiation.

inoue_vicarious_2006; Inoue (2003) notices that in cases where the listener occupies a privileged social position of power over the speaker, they may inhabit a priviliged listening subject position in which the beliefs required to uphold this social hierarchy may play a major role in their construal of that speaker's indexical behavior and override or erase the speaker's intentions. Studying the emergence

of the notion of 'women's language' in Japan, Inoue notices that to Meiji period elite male intellectuals, schoolgirls inhabited a social position which represented a modernizing breach from traditional Japanese social structure, and a threat to their power. For this reason, when they overheard schoolgirls using the verb ending forms *teyo* and *dawa*, they construed them as indexical of degeneracy and social decline. Of course, schoolgirls themselves did not intend to use these forms to index a 'degenerate' identity, and the preported degeneracy these forms represented was a fictional construction, rather than any particular actions these girls were involved with. In this way, male intellectual overhearers constructed a listening subject position which made negative judgemental conjectures about the indexicality of schoolgirls' speech, and constructed a fictional object for the these speech forms to signal. The male listening subject sets the narrative, erasing schoolgirl speakers' semiotic agency.

Following this important insight, research has extended theorization of the listening subject to the questions of race and queerness, researching white (Flores and Rosa 2015) and cisgender listening subjects (**calder_whose_2019**; Calder and Steele 2025; Heffernan 2023; Jessee and Calder 2025). Flores and Rosa (2015) find that when white listeners racialize speakers as non-white, they justify their position of power over those speakers by construing those speakers speech as indexing unspecified 'inappropriateness', regardless of those speakers' adherence to linguistic norms for standard English, constructing a white-supremacist white listening subject which will always deem non-white speech deficient and indexical of non-white inferiority. Jessee and Calder (2025) find that cisgender listeners apply stricter gendered listening standards to an audio clip of a voice when they are told the speaker of the clip is a trans woman than when told the speaker is a cis woman, similarly situating sufficient gender performance out of reach for transgender listeners in a move to deny their identities.

Drawing on these concepts of white, cis, and male listening subjects, **calder_whose_2019; cal** raises an important critique of the sociolinguistic project, questioning which listening subjects researchers themselves may be inhabiting in their interpretation of speakers' language use. Following the conjecturing model of Gal and Irvine (2019), they note that researchers collect numerical correlation data between linguistic forms and social aspects, and use this as the knowledge on which to base their conjectures of indexical links. Incorporating the research on the listening subject, they conclude that although this data may seem objective, emperical, and scientific, purely describing the world as it is (see also Zimman 2021), listening practices inevitably are biased by the sociocultural positioning of the researcher and the research that researcher builds upon. This research raises the question of how to research the social-indexical performativity of language-in-use without taking up and reinforcing violent listening practices and marginalizing speakers who are not white, cis, heterosexual men. Work which addresses this question employs advanced models of indexicality which encourage the object of sociolinguistic study to move from simply asking what meanings participants index to investigating the socio-cultural structure of individuals' indexical knowledge, and the intersubjective process of indexicality (**barett_sexuality_2025**; Calder 2021; Silverstein 2003;

Silverstein 2023). The concepts of the indexical field as adapted by Calder (2021), and indexical (mis)recognition as developed by **barett_sexuality_2025** have been used to model indexical knowledge and the relativity and intersubjectivity of indexicality.

The concept of the indexical field, introduced by Eckert (2008), has been adopted by Calder (2021) to capture the way in which a particular phenomenon may be conjectured to have different meanings by different groups of people. In Gal and Irvine's (2019) semiotic model, conjectures are made based upon listeners' knowledge. In this way, a particular linguistic form may be construed as a sign of any number of meanings, depending on which knowledge the listener uses to conjecture the relationship, and what conclusion they come to in conjecturing. Although this knowledge is mediated and constructed differently by each individual's social and cultural lived experience, it is nonetheless to a large extent shared. For example, Inoue's (2003) listening subject was not a single Meiji-era Japanese male intellectual, it was an ideological position constructed and inhabited by a this male intellectual social class. Individuals who inhabit similar social and cultural backgrounds and social positions, as well as co-members of communities, are likely to draw upon a similar base of knowledge, and are likely to consider certain phenomena signs with a particular set of possible meanings. Calder (2021) conceptualizes the indexical field as this set of meanings which may be shared by a set of people with shared epistemology. By making explicit whose indexical field researchers are investigating and interpreting signs through, the fact that this interpretation is coming from people's perspective is forefronted, preventing the universalization of the hegemonic indexical field (the white, male, heterosexual, cisgender, middle class, etc. indexical field). While this makes progress towards a more critical and less biased move to interpreting signs, and much research has been productively done to investigate the indexical fields within particular local communities, especially community-specific systems of categorization and meaning, it is nonetheless untenable to simply split the world into non-overlapping spheres of shared knowledge and shared indexicality. In other words, indexical fields overlap, and individuals always exist in multiple spaces and must be aware of how indexical conjectures may be made about their behavior across all indexical fields those who they interact with are familiar with. The question that arises is: what happens when a speaker and listener do not share an indexical field?

Barrett and Hall (2024) capture this with their concepts of the indexical inventory and indexical (mis)alignment/(mis)recognition. For them, individual's indexical knowledge is structured in an indexical inventory, which is "an individual's full set of indexical fields." (p. 132) When individuals with different indexical inventories interact and they share similar indexical inventories and indexical fields, they are said to be in indexical alignment. In this case, if a speaker use a linguistic form that they know to hold a particular indexical meaning within one of the indexical fields in their inventory, their listener would recognize (in Barrett and Hall's term) their indexical behavior as coming from the indexical field on which they are aligned, interpreting it in line with the speaker's intentions. However, in cases of indexical misalignment, the speaker employs a linguistic form with the

intent to index a particular meaning present in the indexical field for that form in their inventory, but the listener does not have that indexical field present in their inventory; there are two possibilities at this point. If the listener has no indexical field in their inventory for that form, they may simply not conjecture about it, not knowing that it was intended to be a sign. If they do have an indexial field for that form in their inventory, but it is not the same as the one the speaker intended to call upon, they may conjecture that the sign means something other than what the speaker had intended. They cite the queer "Friend of Dorothy" social routine as an example of a sign which makes purposeful use of indexical misalignment. When a queer person asks "are you a friend of Dorothy?", they may be drawing on an indexical field specific to the queer community in-group in which the meaning of this question is, more or less, "are you gay?" The usefulness of the question comes from the exclusivity of its indexical field; asking a straight person whether they are gay straightforwardly may result in violence, but if it is assumed that straight people are unaware of the routine, but that gay people are aware, then it will only be recognized by gay people, removing the risk (Barrett and Hall 2024, p. 129-30). This model complicates the relationship between the speaker and the listener in indexicality significantly, and raises the primary theoretical question this thesis aims to shed insight on: how do speakers model the indexical inventory of those who may be listening to them, and how do these models influence their behavior to help them achieve their indexical goals, whether those goals be alignment and recognition or misalignment and misrecognition?

I argue that this work of modeling the listener's indexicality and adapting to it is not distributed equally, primarily falling to marginalized speakers (especially multiply-marginalized speakers) due to the power dynamics which govern the construction of indexical fields. All speakers are under coersion to understand the white, male, heterosexual, cisgender, middle class hegemonic indexical field, risking negative judgements from the listening subject position associated with that field if they do not learn its norms and store them in their indexical inventories. However, those who inhabit privileged positions are unlikely to face any negative consequences due to ignorance of indexical fields belonging to marginalized groups, as marginalized people do not have the coercive power to construct a judging listening subject position. For this reason, the investigation of marginalized speakers' modeling of and reaction in behavior to listeners who hold privilege over them emerges as the most viable avenue for the research of this dynamic generally. It is for this reason, as well as further reasons detailed below, that I choose to focus on transfeminine people for my research.

Heffernan (2023) finds that transfeminine speakers are aware of the cisgender listening subject. When asked questions about how they use their voice and how they feel about their voice, transfeminine speakers in Heffernan's study noted that they fear negative judgements by cis people when linguistic gender norms are not followed. Further, her speakers expressed varied ways of navigating this, from accomodating these expectations by a change in linguistic behavior towards conformance to asserting gender to be self-determined, resisting these judgements, and talking as one would prefer. One of the transfeminine speakers in Heffernan

(2023) noted that when "interacting with the cis universe" she finds herself "deliberately modulating [her] voice" to make it "more girly", and that this is "a kind of survival tactic... to try and avoid any possibility that [she]'ll get misgendered" (23). However, this speaker, as well as her conversation partner, also note that "it's kind of just always incorrect to assume someone's gender", and expressed uncertainty about whether she would want to train her voice to be more hegemonically feminine, as she wants to be able to "[think] about (.) [her] voice ... through uh-like the lens of other trans people", who would not apply the hegemonic indexical field, not assume her gender, and not require this sort of modulation. In this way, the decision whether to shift one's speech in the presence of a privileged listener who may inhabit a judgemental listening subject position is complicated and personal. Experimental investigation of this sort of shifting, comparing transfeminine speakers' indexical behavior when talking to a cisgender person with their indexical behavior when talking to a transgender person has not been conducted.

The work of Becker et al. (2022), Gratton (2016), and Parnell-Mooney (2019) represent the nearest examples. Becker et al. (2022) find that trans men and AFAB nonbinary people on testosterone use significantly less creaky voice (known popularly as 'vocal fry') in a reading-passage task compared to in a open-ended conversation. As creaky voice is hegemonically associated with femininity, they tentatively interpret these findings as evidence that in a more scrutinized form of speech (in which, I add, the notion of the cisgender listening subject may be more mentally present), these individuals purposefully move away from normative feminine speech in order to assert their identities as non-women. This aligns with the findings in a study on a Toronto nonbinary community conducted by Gratton (2016). Gratton finds that variation between *-ing* and *in'* (eg. *working* vs. *workin'*) was more even in queer-safe spaces, and more differentiated between the two in public. This was interpreted as in line with their nonbinary identities, as the *-ing* variant is hegemonically associated with femininity and *in'* with femininity. In public, speakers avoided the variant more associated with their assigned gender at birth. That cisgender overhearers may be present in the public setting brought the cisgender listening subject's influence to speakers' behavior; the cisgender listener is invested in fitting them into the binary gender they were assigned at birth, so speakers exaggerated their difference from that gender. In in queer-safe spaces, the speakers felt free to speak more neutrally and without regard for binary norms. Further, ons speaker even noted felling comfortable making occasional use of a 'manly voice' in these spaces for comedic effect despite the danger of being misgendered as a man as a nonbinary individual assigned male at birth; they knew they would not face misgendering for doing so in the absence of cisgender listeners. Parnell-Mooney (2019) obtains similar findings, seeing that transmasculine speakers and an AMAB nonbinary speaker shift further from the use of a form associated with the opposite gender, but associated with their assigned gender at birth, when interacting a cisgender person of that gender than when interacting with a cisgender person of the same gender. Although these studies present important emperical results, they do not integrate with the theoretical questions of indexical recognition, indexical fields, and the modeling of a listener's indexical

knowledge raised above, do not include transfeminine speakers, and to not make a direct comparsion between conversation between a cisgender listener and conversation between a transgender listener. Filling this gap, I compare transfeminine speakers' indexical behavior across a conversation with a cisgender woman and a transgender woman with reference to their commentaries on their own gender, the voice, and being heard as a transfemine speaker.

For my comparison, I focus specifically on the /s/ sound, as the first sound in 'safe' or the last sound in 'face', due to the richness of existing research on its indexicalities and its use by transgender individuals, especially in connection to the sophisticated theories of indexicality discussed above. The remainder of this section introduces this literature on /s/ and reviews its connections to the concerns of transgender language and indexicality research.

1.2 The Trans Sociophonetics of /s/

1.3 disjointed stuff

One approach to this problem is the notion of indexical orders developed by Silverstein (2003; 2023). As linguistic forms are construed as indexical signs and gain meanings within a particular indexical field, it is possible that new meanings will arise which build off of and are informed by the original meaning. This can take place both within a single indexical field, and cross-cutting the overlaps of multiple. It is helpful here to

For example, a particular linguistic form may come to index femininity in the hegemonic indexical field. This semiotic development would already exist informed by an innumerably long chain of previous semiotic knowledge and material, and for this reason is said to exist at the nth indexical order (Silverstein 2003). However, when that form is used by those percieved as men, the combination of these two significations brings the interpretation to a new indexical order – not only is femininity indexed, but gay masculinity often arises as an additional meaning at the $n+1$st order. These meanings, situated at different levels, inform and construct each other, and do not form a chronological order despite their ordered structure. This development of the gay masculinity index at the $n+1$st order may also occupy the hegemonic indexical field, especially if users of this form are stigmatized if they are percieved as men. Yet, further indexical orders can develop that are situated within other indexical fields. For example, if one group of drag queens uses this form differently than another, an indexicality at the $n+1+1$nd order may emerge whereby those in that drag community may understand the form to index membership in a particular drag group.

calder_whose_2019

By making claims about the indexical meaning of linguistic forms, scholars are conjecturing based on their socioculturally mediated knowledge.

Although these scholars do not make value judgements of these speakers, de-